Rico Trebeljahr



Software Engineering Consultant

System recovery / Architecture / API integrations / Delivery automation

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German: native / English: C1

Software engineer and consultant for inherited systems, architecture decisions and API integrations. Combines technical investigation with implementation, tests, deployment automation and handover.

SELECTED CONSULTING AND ENGINEERING WORK

System recovery - Henkel: took over a poorly documented Azure pipeline processing 100,000+ events/day after a two-week handover. Mapped data flows, added unit/E2E tests and documented the system for two incoming engineers.

Architecture decisions - Softgames: compared six multiplayer-backend options against scalability, cost and team capability. Presented the CTO-approved Go/Nakama recommendation and led backend implementation in a two-person team.

API integrations: built third-party KYC integrations for a confidential client across three engagements. In Track Your Time, designed REST/OpenAPI contracts with scoped tokens and structured errors.

Delivery automation - Hatchkit: built reusable TypeScript setup and deployment tooling; applied shared configuration across 12 other own product repositories. Used Hatchkit for Track Your Time's initial working cross-platform prototype in under one working week.

Independent portfolio: Fractal Garden (interactive fractal and mathematics website); Mesozoic Protocol (3D tower defense game); Tiao (multiplayer board-game platform)

SELECTED CLIENT EXPERIENCE

Confidential client · KYC integrations (freelance) — Dec 2025, Apr/Jul 2026

Henkel Dx · IT Consultant (freelance) — Apr 2023 - Nov 2023

European Space Agency · Full-stack Engineer (via IMS) — Oct 2021 - Dec 2021

Klarna · Software Engineer II — Jan 2021 - Jun 2021

Full chronology and project detail follow.

CORE TECHNOLOGIES

Frontend	TypeScript, React, Next.js, TanStack Query
APIs & data	Node.js, REST/OpenAPI, MongoDB, Redis, WebSockets
Delivery	Git (2021-present), CI/CD, Docker, Terraform, Azure

Professional experience

Confidential client

Dec 2025; Apr 2026; Jul 2026 | Freelance software development / Repeat engagements

- Built third-party integrations for Know Your Customer (KYC) workflows.
- Three separate engagements: one month in December 2025, one month in April 2026 and one week in July 2026.

flowkey

Oct 2025 | IT Consultant / Freelance

- Implemented cookie-consent designs across the company website and web application.
- Worked within the existing Next.js and React product environment.

Technologies: TypeScript, React, Next.js

Henkel Dx

Apr 2023 - Nov 2023 | IT Consultant / Freelance / Sole developer

- Maintained an inherited customer-data pipeline processing 100,000+ events/day as sole developer, following a two-week handover.
- Mapped system hierarchy and data flows; introduced unit/E2E tests and mock factories to support refactoring.
- Proposed consolidation from about 150 Azure Functions and six service buses to six functions and one bus. Handed over before production rollout.
- Integrated SAP, Facebook data enrichment and Adobe Experience Platform; delivered integration changes, documentation and handover to two replacement engineers.

Technologies: TypeScript, Microsoft Azure Functions, Azure Service Bus, SAP, Facebook Marketing API, Adobe Experience Platform

Ironhack

Jan 2022 - Jun 2023 | Web Development Lead Teacher / Freelance

- Led three cohorts covering HTML, CSS, JavaScript, MongoDB, Express, React and Node.js.
- Taught 50+ career changers across two nine-week intensive programs and one six-month part-time program.
- Guided students through building and deploying full-stack applications.

Technologies: JavaScript, HTML, CSS, React, Node.js, Express, MongoDB

Earlier professional experience

European Space Agency / Space Debris Office

Oct 2021 - Dec 2021 | Full-stack Engineer / Engagement via IMS

- Built software supporting satellite collision-avoidance workflows.
- Unified orbit-data conversion across command-line tools and multiple orbit-file formats.
- Automated deployments with Ansible and contributed to CI/CD and testing improvements.

Technologies: Python, Flask, SQLAlchemy, React, Ansible

Klarna

Jan 2021 - Jun 2021 | Software Engineer II / Browser extension team

- Built automated end-to-end testing infrastructure for the Chrome extension using Cypress.
- Integrated tests into CI/CD and implemented the extension notification feature.

Technologies: TypeScript, React, Cypress, Chrome Extension APIs, CI/CD

Softgames

Jun 2019 - Apr 2020 | Junior Software Engineer / Multiplayer backend

- Compared Nakama, Photon, PlayFab, GameSparks, Colyseus and a custom backend for a Snapchat multiplayer quiz game.
- Balanced horizontal scaling, cost, team expertise, matchmaking/netcode effort and architecture fit. Presented a Go/Nakama recommendation; the CTO approved the choice.
- Led implementation of matchmaking and synchronized game state; onboarded a senior engineer into the two-person backend team.
- Built a TypeScript/RxJS client library for frontend data requirements.

Technologies: Go, Nakama, TypeScript, RxJS, Google Cloud Platform

Product engineering and ownership

Ricos Labs LLC: Apr 2026 - Present. Independent product development: Jan 2023 - Present. Selected projects below are independent products, not client engagements.

Tiao

2026 - Present | Digital implementation / Full-stack product engineering

- Made a physical board game playable remotely in the browser, with shared rules and an interactive tutorial.
- Shipped the initial online implementation in two weeks using AI-assisted development; continued to develop the product afterwards.
- Implemented WebSocket multiplayer, matchmaking, tournaments, friends, game history, ratings and an interactive tutorial.
- Built shared TypeScript game rules for client and server validation, plus an in-browser computer opponent.
- Integrated email/password and OAuth authentication, payments, storage and deployment workflows.
- Implemented Redis-backed coordination for distributed matchmaking, locks and rate limiting.
- Delivered the digital platform for the board game invented by Andreas Edmeier.

Technologies: TypeScript, React, Next.js, Node.js, Express, WebSockets, MongoDB, Redis, Docker, Coolify, Terraform, GitHub Actions

playtiao.com

Track Your Time

2026 - Present | Time tracking, invoicing and multi-client product

- Reached a working prototype in under one working week using Hatchkit: core workflow, authentication, deployed web and desktop/mobile builds. UI polish and feature expansion followed.
- Built a shared Express/tRPC backend and Next.js client for time tracking, reports, budgets, teams and invoices.
- Implemented invoice creation from server-recomputed billable entries, snapshotted rates and billing identities, sequential numbering and protections for invoiced time.
- Added ZUGFeRD and XRechnung export with validator-backed CI samples.
- Implemented TanStack Query caching, query persistence, cache invalidation and sign-out cleanup across web and native clients.
- Designed REST contracts with scoped tokens, cursor pagination and RFC 9457 errors; generated OpenAPI 3.1 from shared route schemas.
- Built account- and workspace-scoped offline mutation queues and authenticated WebSocket synchronization.
- Implemented workspace roles, visibility rules, device authorization and revocable sessions across clients.
- Delivered desktop downloads for macOS, Windows and Linux and a Chrome extension; iOS and Android builds are in testing, not public app-store releases.

Technologies: TypeScript, Next.js, Express, tRPC, MongoDB, Redis, Electron, Capacitor, WebSockets, REST API, MCP, TanStack Query, OpenAPI 3.1, Git

trackyourtime.dev

Developer tools and delivery automation

Hatchkit

2026 - Present | Project scaffolding and infrastructure CLI

- Applied shared project setup and service configuration across 12 other own product repositories.
- Built a full-stack TypeScript CLI/starter with shared client/server contracts and optional auth, WebSockets, payments, storage and cross-platform shells.
- Automated Terraform/Hetzner, Coolify, DNS and Cloudflare Workers deployment; separated frontend hosting from the backend runtime.
- Used one integrated starter and removed unselected features to reduce integration duplication. Added ML-service deployment, observability and Listmonk/AWS SES email setup.
- Built signing and release workflow setup for desktop and mobile targets, plus an on-demand local macOS CI runner.
- Provided interactive prompts, scriptable flags, diagnostics and read-only MCP tools for operators and coding agents.

Technologies: TypeScript, Node.js, CLI, Terraform, Docker, Hetzner, Coolify, GitHub Actions, Electron, Capacitor, Cloudflare Workers, Listmonk, AWS SES, MCP

hatchkit.trebeljahr.com

sprite-tools

2026 - Present | Sprite-processing pipeline / Web app, CLI and MCP server

- Built a shared processing core exposed through three interfaces: browser tools, a command-line application and a Model Context Protocol server.
- Implemented sheet inspection, background removal, frame deduplication, collision polygons, pivot metadata, palette transforms and atlas packing.
- Added Aseprite import and exports for Godot, Unity, Aseprite and Phaser, preserving animation metadata.
- Kept local image processing in the browser; optional provider-backed generation is a separate explicit flow.
- Made pipeline output composable through common JSON shapes across the three interfaces.

Technologies: TypeScript, Next.js, Canvas, Node.js, CLI, MCP, Vitest

sprites.trebeljahr.com

conv3d

2023; maintained in 2026 | Published npm CLI / 3D asset conversion

- Built batch conversion from FBX, OBJ and glTF to GLB with optional React Three Fiber component generation.
- Added interactive and non-interactive operation, dry runs, JSON output and configurable conversion concurrency.
- Supported files, directories and glob patterns for repeated web-asset preparation.

Technologies: TypeScript, Node.js, glTF, GLB, gltfjsx, CLI

conv3d.trebeljahr.com

Interactive graphics and audience reach

Independent browser-based graphics and mathematical exploration.

Fractal Garden

2022 - 2026 | Interactive mathematical exhibition

- Recorded 9,000+ Google Search clicks and about 184,000 impressions from 20 Jul to 19 Oct 2026 (Google Search Console).
- Averaged Google positions 2.3 for "burning ship fractal" and 4.6 for "mandelbrot set generator" over that period.
- Reached about 5,700 recorded visitors in the 28-day Plausible snapshot checked on 22 Oct 2026.
- Built an interactive collection of 53 mathematical exhibits spanning complex-plane shaders, L-systems, 3D fractals and procedural patterns.
- Implemented GLSL rendering and interactive controls to explore mathematical behavior in the browser.
- Expanded and maintained the exhibition across successive development periods.

Technologies: React, Next.js, WebGL, GLSL, Canvas, L-systems

fractal.garden

ricos.site / Interactive 3D gallery

2021 - Present | Publishing platform and graphics experiments

- Built and maintained a custom publishing platform with MDX content, full-text search, image handling and newsletter infrastructure.
- Built and adapted 40+ interactive React Three Fiber and GLSL demos, including procedural terrain, particles and shader experiments.
- Integrated interactive explanations and graphics into a larger content platform.
- Maintained the publishing repository with recurring Git history from 2021 through 2026.

Technologies: TypeScript, React, Next.js, MDX, React Three Fiber, Three.js, GLSL, Git

ricos.site/r3f

Games and platform delivery

Solo development; playable builds verified on iOS, Android, Linux and macOS. Commercial store launches are pending.

Mesozoic Protocol

2026 - Present | 3D tower-defense game / In development

- Built a headless TypeScript simulation with a React Three Fiber renderer and persistent campaign progression.
- Implemented a campaign with 30 outposts, six tower types and multiple biomes, plus upgrade and targeting systems.
- Developed web, Tauri desktop and Capacitor mobile build targets; distribution and store work continue separately.

Technologies: TypeScript, React Three Fiber, Three.js, Zustand, Vite, Tauri, Capacitor

mesozoicprotocol.com

Raptor Runner

2026 - Present | Cross-platform Canvas game

- Built a Canvas game with day/night cycles, weather, collision detection, cosmetics and React interface overlays.
- Used one TypeScript codebase for the web, Electron desktop and Capacitor mobile builds.
- Maintained build, test and release workflows across the platform targets.

Technologies: TypeScript, Canvas 2D, React, Vite, Electron, Capacitor, Vitest

raptorrunner.com

Technical capabilities and foundations

Full-stack application engineering

2019 - Present | Selected applied technologies

- Git: recurring version-control use in maintained repositories across 2021-2026; release history and CI/CD workflows.
- Frontend: TypeScript, JavaScript, React and Next.js; web interfaces, browser extensions and desktop/mobile shells.
- TanStack Query: cache lifetime, persistence, invalidation and account/workspace isolation in Track Your Time (2026).
- Backend: Node.js, Express and tRPC in recent products; Python/Flask at ESA and Go/Nakama at Softgames.
- Data and real-time systems: MongoDB and Redis in recent products; SQLAlchemy in client work; WebSocket state synchronization and workspace-scoped authorization.
- Quality and delivery: Cypress, Vitest, CI/CD, Docker, Terraform and Ansible across client work and independent products.
- Graphics and assets: Three.js, React Three Fiber, GLSL, Canvas, procedural generation, glTF/GLB and sprite-processing workflows.
- AI-assisted engineering: used coding agents to accelerate implementation while retaining architecture, testing, review and release responsibilities.

Additional engineering projects

2019 - 2026 | Selected foundations

- Chess: full-stack multiplayer project, originally built with Meteor and later modernized with Next.js, Express, Socket.io and MongoDB.
- Minecraft Clone: browser voxel world using procedural biome and cave generation, block interaction and custom chunk rendering.
- Quaternius Showcase: an interactive browser viewer for third-party 3D assets; the models are by Quaternius.

Technologies: TypeScript, React, Three.js, WebSockets, Node.js

portfolio.trebeljahr.com

Education and continuing study

2019 - Present | Self-directed education / No university degree

- Studied web development through freeCodeCamp before the first professional software role in 2019.
- Self-studied calculus, differential equations and linear algebra through MIT OpenCourseWare, plus online machine-learning courses.
- Applied learning through commercial delivery, technical instruction and independently maintained products. MIT materials are self-study, not an MIT degree or credential.

Engagement focus

Current | Freelance software development and technical consulting

- German: native / English: C1
- Full-stack product delivery, real-time applications, developer tooling and interactive graphics.
- Project-based work with defined scope and milestones; contract terms depend on the client and delivery requirements.

ricoslabs.com